



LESSON 6.11

REASONING THROUGH PROBLEMS INVOLVING PROPORTIONAL RELATIONSHIPS

LESSON INTRODUCTION

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

MA.7.AR.4.5 Solve real-world problems involving proportional relationships.

LEARNING TARGETS

- I can solve multi-step problems involving proportions.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- How can you use patterns/relationships from the problem to solve the problem?
- How can you use a proportional relationship to solve a multi-step ratio problem?

LESSON NARRATIVE

This lesson is designed primarily as a day for students to practice and apply what they have learned throughout the unit. The lesson begins with an error analysis Collaboration, after which students summarize some of the helpful tips and strategies they have picked up throughout the unit while solving multi-step problems involving proportional thinking. This is followed by additional student practice with various contexts.

COMPONENT SUMMARY

6.11.1 WARM-UP (~5 MIN.)

Video: Paint Mixing

6.11.3 BRING IT TOGETHER (10 MIN.)

Create a list of tips and strategies that can be used to solve proportion problems

6.11.5 WRAP-UP (~5 MIN.)

Emoji scale reflection as students conclude the unit

6.11.2 COLLABORATION (10 MIN.)

Engage in an error analysis to explain and correct a given student's work

6.11.4 YOUR TURN! (10-15 MIN.)

Practice creating and solving proportions to answer six problems

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- 6.11.1 Warm-Up video

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle with differentiating between part-to-part vs part-to-whole comparisons in context.
- Students may try to use addition or subtraction instead of multiplication or division to find the scale factor.
- Students may struggle with division of fractions within a ratio.
- Students may think that every part of their work shown needs a proportion or rate.

THINGS TO CONSIDER

- This lesson focuses less on the structure of a proportion or ratio and more on how the proportion provides information that can answer the problem question.
- There are a variety of strategies students have learned throughout the unit that can be applied to solve these problems. Consider that there is more than one way to solve the problem that shows students understand the relationships.

LITERACY AND LANGUAGE ROUTINES

Collect and Display

As students reflect on proportional problem-solving in 6.11.2 Collaboration and 6.11.3 Bring It Together and as students work in 6.11.4 Your Turn!, the cocreation of and visible access to the references will remind students of ways to overcome any obstacles in proportional problem-solving, keeping the pace of the lesson moving forward and solidifying understanding through habitual practice.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Consider prompting students to assess the reasonableness of each solution in 6.11.4 Your Turn! by having them write a verbal description of their final answer. This strategy builds reflection while also acting as a final check for errors in their work.

DIFFERENTIATE INSTRUCTION

In 6.11.2 Collaboration and 6.11.3 Bring It Together, provide a stable reference that students build for the entire class to see and create the space for whole-class support. Monitor as pairs work to select which work to write on the poster.

6.11.3 Bring It Together allows flexibility for students to make sense of other students' thinking in the ways that make most sense to them, whether that be hearing it, drawing it, or writing it out.

During 6.11.5 Wrap-Up, some students may be hesitant to rate themselves accurately and may voice that their rating would be different on different days. Offer students the option to draw a range and label dates or kinds of problems to justify their differences in ratings.

6.11.1 WARM-UP: MIXING PAINT

TEACHER MOVES

Monitor the class for students who got 24 parts and have them share their reasoning. Students who got 24 as the answer likely realized that Dan would have already included 1 part when solving the proportion, the first time around, so only 24 more parts are needed.



6.11.1 Warm-Up: Mixing Paint

Dan is helping his Nana to make the paint color she needs. He mixes 1 part red paint with 5 parts white paint, as she requested. Nana realized she gave him the information backwards: It was supposed to be 5 parts red paint and only 1 part white paint.

- How can Dan fix it?

$$\frac{\text{red}}{\text{white}} = \frac{1}{5} = \frac{?}{5}$$

25 parts red but already has 1...

So

add 24 parts red paint.



6.11.2 Collaboration: Error Analysis

Chloe's phone has 64 gigabytes (GB) of memory. One GB is equal to 1,000 megabytes (MB). She takes a lot of pictures, regularly streams videos, and typically saves all of her messages. She reserves about 50 GB for those purposes. If the average app uses about 200 MB of memory, how many apps would she be able to download to her phone?

$$50 \cancel{\text{GB}} \times \frac{1000 \text{ MB}}{1 \cancel{\text{GB}}} = 50,000$$

$$\frac{1 \text{ app}}{50,000 \text{ MB}} = \frac{a}{200 \text{ MB}} \quad a = 250 \text{ apps}$$

1. With your partner, discuss the errors that Chloe made. List the errors.

- She set up the proportion wrong
- She found how many apps for 50 GB instead of remaining 14 GB

2. Help Chloe understand and learn from her mistakes. Write a brief note to Chloe explaining what her mistakes were and how she could learn from them.

Answers will vary. Good job Chloe on setting up a proportion. However, you found the apps for 50 GB instead of 14 GB. Underline the important phrases and circle the important numbers in the word problem prior to setting up the proportion. In addition, check your final answer to ensure it makes sense for the problem.

3. Work with your partner to determine how many apps Chloe would be able to download to her phone.

$$\cancel{14 \text{ GB}} \times \frac{1000 \text{ MB}}{1 \cancel{\text{GB}}} = 14,000 \text{ MB}$$

$$\frac{1 \text{ app}}{200 \text{ GB}} = \frac{a}{14,000 \text{ MB}} \quad a = 70 \text{ apps}$$



6.11.3 Bring It Together: Solving Multi-Step Problems

Consider some of the problems you've solved and errors you've made over the last few lessons.

1. What are some strategies you have learned and key tips that are helpful for you to keep in mind when solving multi-step problems?

Helpful Tips	Strategies
<i>Answers will vary.</i> • Underline key words. • Circle key amounts. • Double check your multiplication. • Ensure that you are using the right conversion rates. • Ensure proportions are set up correctly based on each context.	<i>Answers will vary.</i> • Check the answer to your proportion to ensure it makes sense in the context. • Cross out units as they are eliminated.

2. Compare and discuss your list with your partner. Add any tips or strategies your partner shares to your list that you find helpful.

6.11.2 COLLABORATION: ERROR ANALYSIS

ENRICHMENT

Based on the context, what is wrong with the work Chloe showed?

TEACHER MOVES

Collect and Display

Consider projecting the student work on a board. Prompt students to share their responses and capture both their verbal and algebraic descriptions, making use of markers and colors to make visual connections for students.

QUESTIONING

Ask those who finish early to find other ways to show their work for this problem.

6.11.3 BRING IT TOGETHER: SOLVING MULTI-STEP PROBLEMS

DIFFERENTIATE INSTRUCTION

If students are confused about the distinction between tips and strategies, describe them as such:

Tip: What action did you do to understand?

Strategy: What method or visual cue could you use to represent how you are solving?

ENRICHMENT

What have you used in the previous lessons that helped you to understand and solve the problems?

6.11.4 YOUR TURN!

DIFFERENTIATE INSTRUCTION

Encourage students to use one tip and one strategy from the previous activity to help them solve these problems if they get stuck or have them swap tips and strategies with a partner and use their partner's tips.

ENRICHMENT

- What is the total perimeter of the backyard in inches?
- What is the total perimeter of the backyard in feet?

ENRICHMENT

Is this a part-to-part or part-to-whole proportion problem?

TEACHER MOVES

Look for students who do not correctly apply conversions between meters and feet.

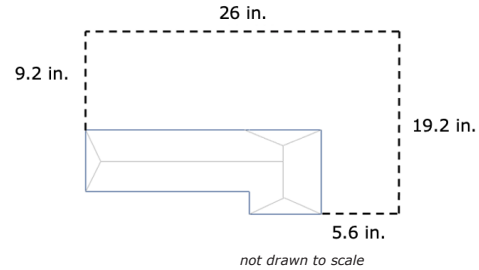
TEACHER MOVES

Look for students who incorrectly set up a ratio that compares 1.4 miles to 3.2 miles per hour. Prompt students to use the units in setting up the proportion.



6.11.4 Your Turn!

- Valerie is putting in a new fence in her backyard. A scale drawing of her backyard is shown, with the dashed line representing where the fence will be installed. The scale used for the drawing is 2.5 inches = 7 feet



$$P = 9.2 \text{ in} + 26 \text{ in} + 19.2 \text{ in} + 5.6 \text{ in} = 60 \text{ in}$$

The fence she wants to install costs about \$17.25 per linear foot. How much will her new fence cost?

$$60 \text{ in} \times \frac{7 \text{ ft}}{2.5 \text{ in}} \times \frac{\$17.25}{1 \text{ ft}} = \$2,898$$

- A painter mixes paint to create a specific shade of purple. The ratio of each color in the mixture is 1 part blue paint, 3 parts red paint, and 4 parts white paint. The painter makes a large batch of purple paint. He uses 6 gallons (gal) of white paint in the mixture. How many total gallons is the batch of paint he made?

$$\frac{\text{white}}{\text{total}} \Big| \frac{4}{8} = \frac{6}{?} \quad 12 \text{ gallons}$$

- The largest U.S. flag in the world is 225 feet (ft) by 505 feet. If one square meter of the flag weighs about 3.8 ounces, how much does the entire flag weigh to the nearest tenth of an ounce?

$$1 \text{ square meter} = 10.76 \text{ square feet}$$

$$225 \times 505 = 113,625 \text{ ft}^2$$

$$113,625 \text{ ft}^2 \times \frac{1 \text{ m}^2}{10.76 \text{ ft}^2} \times \frac{3.8 \text{ oz}}{1 \text{ m}^2} = 40,127.8 \text{ oz.}$$

- Paul has determined that the number of calories in a 2-scoop serving of ice cream is 375 calories. Paul has $1\frac{1}{2}$ scoops of ice cream 4 days a week. How many calories from ice cream does Paul eat in a week?

$$1\frac{1}{2} \text{ scoops} \times \frac{375 \text{ cal}}{2 \text{ scoops}} \times \frac{4 \text{ days}}{1 \text{ week}} = 1,125 \text{ calories}$$

- Maylisa averages 3,247 steps for 1.4 miles (mi). She can walk 3.2 miles in an hour. How many steps can Maylisa walk in 90 minutes?

$$90 \text{ min} \times \frac{1 \text{ hr}}{60 \text{ min}} = 1.5 \text{ hrs}$$

$$\frac{1 \text{ hr}}{3.2 \text{ mi}} = \frac{1.5 \text{ hrs}}{d} \quad \therefore d = \frac{1.5 \text{ hrs} \times 3.2 \text{ mi}}{1 \text{ hr}} = 4.8 \text{ mi}$$

$$\frac{1.4 \text{ mi}}{4.8 \text{ mi}} = \frac{3247 \text{ steps}}{n} \quad \therefore n = \frac{3247 \text{ steps} \times 4.8 \text{ mi}}{1.4 \text{ mi}}$$

$$= 11,132.57 \text{ steps}$$

$$11,132.6 \text{ steps or } 11,133 \text{ steps}$$

6. Rishi and Diego are each making orange juice. Rishi mixes $2\frac{1}{2}$ cups (c) of water with $\frac{1}{3}$ cup of orange juice concentrate. Diego mixes $1\frac{2}{3}$ cups of water with $\frac{1}{4}$ cup of orange juice concentrate.

- a. Will their orange juices taste the same? Explain.

$$\frac{\frac{1}{3}}{2\frac{1}{2}} = \frac{\frac{1}{4}}{1\frac{2}{3}} \div \frac{2}{15} = \frac{3}{20} \approx 0.15 \neq 0.13$$

No, Diego's orange juice will have more orange juice concentrate than Rishi's.

- b. There are 16 cups in one gallon (gal). How much concentrate should each of them mix with 2 gallons of water to make juice that tastes the same as their original recipe?

$$\text{Rishi: } \frac{\frac{1}{3}}{2\frac{1}{2}} = \frac{?}{32}; \frac{32 \times \frac{1}{3}}{2} = \frac{32}{3} \cdot \frac{2}{5} = \frac{64}{15} \quad 4\frac{4}{15} \text{ cups}$$

$$\text{Diego: } \frac{\frac{1}{4}}{1\frac{2}{3}} = \frac{?}{32}; \frac{32 \times \frac{1}{4}}{\frac{5}{3}} = \frac{32}{4} \cdot \frac{3}{5} = \frac{96}{20} = \frac{24}{5} \quad 4\frac{4}{5} \text{ cups}$$

6.11.4 YOUR TURN! (CONTINUED)

TEACHER MOVES

Collect and Display

As students reach the end of this problem set, pick students to write the answer to a problem's question on a class-wide document or board. Instruct them to explain in words why their answer is correct as a justification, similar to 6.11.2 Collaboration.



6.11.5 Wrap-Up: Reflection

1. Consider the learning target from the last few days and mark where you are on the smiley-face scale.
Answers may vary.

I can solve multi-step problems involving proportions.



0 1 2 3 4 5 6 7 8 9 10

2. Explain why you marked yourself where you did.
Answers may vary.

6.11.5 WRAP-UP: REFLECTION

ENRICHMENT

Consider which specific prompts might gather the information you need as students conclude the unit.

- What influenced your decision?
- What have you been doing well or easily?
- What have you been doing that is difficult?
- What haven't you been doing?
- Have you tried any tips and strategies that seem to work for others?